

Program Summary - Project.sas

Execution Environment

Author: u64519491
 File: /home/u64519491/Project/Project.sas
 SAS Platform: Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
 SAS Host: ODAWS02-APSE1.ODA.SAS.COM
 SAS Version: 9.04.01M8P02222023
 SAS Locale: en_GB
 Submission Time: 25/05/2026, 17:44:22
 Browser Host: 223.185.41.27
 User Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
 Application Server: ODAMID00-APSE1.ODA.SAS.COM

Code: Project.sas

```

/*****
* PROJECT: Clinical Trial Data Standardization (CDISC SDTM Pipeline)
* DOMAINS: Demographics (DM) & Adverse Events (AE)
* TARGET: FDA Submission-Ready Datasets
* BY: Abhiraj Harsha Gohatre
*****/

/*=====
* PHASE 1: THE SUBJECT MASTER TABLE (DEMOGRAPHICS - DM) - Working on Demographics File from CDISC-DATASET-GENER
*=====*/

/*1. Ingest Raw Demographics Data */
proc import datafile="/home/u64519491/Project/raw_dm.csv"
out=Work.raw_dm
dbms=csv
replace;
getnames=YES;
run;

/*2. Audit the structural metadata of the imported dataset */
proc contents data=work.raw_dm;
run;

/*3. map it to strict cdisc SDTM Demographics standard */
data WORK.sdtm;
set work.raw_dm (RENAME=(RFSTDTC=raw_rfstdtc RFENDTC=raw_rfendtc RFXSTDTC=raw_rfxstdtc));
/*A. establish structural armour (length/label)*/
attrib
STUDYID LENGTH=$20 label="Study identifier"
DOMAIN LENGTH=$2 label="Domain Abbreviation"
USUBJID LENGTH=$40 label="Unique Subject Identifier"
SUBJID LENGTH=$10 label="Subject Identifier for Study"
RFSTDTC LENGTH=$10 label="subject reference start date time"
RFENDTC LENGTH=$10 label="subject reference end date time"
RFXSTDTC LENGTH=$10 LABEL="date/time of First Study Treatment"
AGE label="Age"
AGEU LENGTH=$10 label="Age Units"
SEX LENGTH=$1 Label="Sex"
RACE LENGTH=$60 Label="Race"
ETHNIC LENGTH=$60 Label="Ethnic Group"
ARM LENGTH=$60 Label="Description of Planned Arm";
/*B. execute data-type conversion (numeric to ISO8601 Character strings) */
RFSTDTC = PUT(raw_rfstdtc, YYMMDD10.) ;
RFENDTC = PUT(raw_rfendtc, YYMMDD10.) ;
RFXSTDTC = PUT(raw_rfxstdtc, YYMMDD10.);
/*C. clean up workspace by dropping temporary numeric variables */
drop raw_rfstdtc raw_rfendtc raw_rfxstdtc ;
/*D. Audit Final Standardized Dataset structure */
proc contents data=work.sdtm;
run;

/*4. Audit Categorical Values for Controlled Terminology Compliance */
proc freq data=work.sdtm;
tables SEX RACE ETHNIC ARM ARMCD ;
run;

/*5. Lock the Clean Dataset into Final Output Workspace */
data work.dm_final;
set work.sdtm;
SEX = upcase(strip(SEX));
ARMCD = upcase(strip(ARMCD));
IF MISSING(USUBJID) then PUT "WARNING: Missing USUBJID for Observation" _N_;
run;

/*=====
  
```

```

* PHASE 2: THE ADVERSE EVENTS OBSERVED DOMAIN (AE)- Working on ADVERSE EVENT File from CDISC-DATASET-GENERATOR
*=====*/

/*1. Ingest Raw Adverse Events Data */
proc import datafile="/home/u64519491/Project/raw_ae.csv"
out=work.raw_ae dbms=csv replace; getnames=YES;
run;

/*2. Sort AE data */
proc sort data=work.raw_ae out=work.ae_sorted;
by USUBJID AESTDTC AETERM ;
run;

/*3. Map/Structure/Generate SDTM AE Architecture */
data work.ae_final;
set work.ae_sorted (rename=(AESTDTC=raw_aestdtc AEENDTC=raw_aeendtc));
by USUBJID;
attrib
STUDYID length=$20 label="Study Identifier"
DOMAIN length=$2 label="Domain Abbreviation"
USUBJID length=$40 label="Unique Subject Identifier"
AESEQ label="Adverse Event Sequence Number"
AETERM length=$200 label="Reported Term for the Adverse Event"
AEDECOD length=$200 label="Dictionary-Derived Term"
AESTDTC length=$10 label="Start Date/Time of Adverse Event"
AEENDTC length=$10 label="End Date/Time of Adverse Event"
AESEV length=$5 label="Severity"
AESER length=$1 label="Serious Event" ;
/* A. set domain code B. formatting Dates and Terms
C. control terminology D. Logic on Sequence Number*/
DOMAIN = "AE" ;
AEDECOD = upcase(strip(AETERM));
if raw_aestdtc ne . then AESTDTC = put(raw_aestdtc, yymmdd10.) ;
if raw_aeendtc ne . then AEENDTC = put(raw_aeendtc, yymmdd10.) ;
AESEV = upcase(strip(AESEV));
AESER = upcase(strip(AESER));
retain AESEQ;
if first.USUBJID then AESEQ = 1 ;
else AESEQ = AESEQ+1 ;
drop raw_aestdtc raw_aeendtc;
run;

/*=====* *
PHASE 3: REGULATORY QUALITY CONTROL & INTEGRITY AUDIT
*=====*/

/*1. verify Meta-Data Constraints for DM Domain */
proc contents data=work.dm_final;
run;

/*2. Verify Meta-Data Constraints for AE Domain */
proc contents data=work.ae_final;
run;

/*3. Cross-Domain Validation: Confirm all AE patients exist in DM master */
proc sql;
select count(distinct USUBJID) as Orphaned_AE_Patients
from work.ae_final
where USUBJID not in (select distinct USUBJID from work.dm_final);
quit;

/*4. Display Final Standardized Data Samples */
proc print data=work.dm_final (obs=5);
title " SDTM Demographics (DM) Final Compliance View " ;
run;
proc print data=work.ae_final (obs=5);
title " SDTM Adverse Events (AE) Final Compliance View " ;
run;
title;

/*=====*
* PHASE 4: THE SQL RELATIONAL JOIN & SAFETY PROFILE TFL
*=====*/

/* 1. Execute relational join and standardize text directly in the database */
proc sql;
create table work.safety_profile as
select
dm.USUBJID,
dm.AGE,
dm.SEX,
dm.RACE,
dm.ARM,
ae.AESEQ,
ae.AETERM,
ae.AEDECOD,
case

```

```

when missing(ae.AESEV) then "No Adverse Event"
else propcase(strip(ae.AESEV))
end as AESEV length=16 label="Severity/Intensity",
ae.AEREL
from work.dm_final as dm
left join work.ae_final as ae
on dm.USUBJID = ae.USUBJID;
quit;

/* 2. Generate the final, publication-ready Clinical Safety Summary Report */
proc freq data=work.safety_profile;
tables ARM * AESEV / nocol nopercnt norow;
title "Safety Audit: Comprehensive Population Coverage (Including No-AE Subjects)";
run;
title;

/*=====
* PHASE 5: PRODUCTION TFL - CLINICAL DATA LISTING (NEW)
*=====*/

options nodate nonumber orientation=landscape;
ods listing close;

/* Generate Clinical-Grade Subject Adverse Event Listing */
proc report data=work.safety_profile nowd headskip;
where AESEV ne "No Adverse Event";
column ARM USUBJID AESEQ AEDECOD AESEV AEREL;
define ARM / order "Treatment Arm" width=15;
define USUBJID / order "Unique Subject ID" width=25;
define AESEQ / display "AE Seq" center width=8;
define AEDECOD / display "Standardized Term (MedDRA)" width=30;
define AESEV / display "Severity" width=15;
define AEREL / display "Causality" width=15;
break after ARM / skip;
title1 "Protocol:STUDY001";
title2 "Listing 16.2.7: Adverse Events Log - Patients with Documented Events";
run;
title;
options date number orientation=portrait;

```

Log: Project.sas

Warnings (12)

Notes (45)

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70      /*=====
71      * PROJECT: Clinical Trial Data Standardization (CDISC SDTM Pipeline)
72      * DOMAINS: Demographics (DM) & Adverse Events (AE)
73      * TARGET: FDA Submission-Ready Datasets
74      *=====*/
75
76      /*=====
77      * PHASE 1: THE SUBJECT MASTER TABLE (DEMOGRAPHICS - DM) - Working on Demographics File from CDISC-DATASET-GENERATOR
78      *=====*/
79
80      /*1. Ingest Raw Demographics Data */
81      proc import datafile="/home/u64519491/Project/raw_dm.csv"
82      out=Work.raw_dm
83      dbms=csv
84      replace;
85      getnames=YES;
86      run;

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to
WORK.PARMS.PARMS.SLIST.
87      /*=====
88      * PRODUCT: SAS
89      * VERSION: 9.4
90      * CREATOR: External File Interface
91      * DATE: 25MAY26
92      * DESC: Generated SAS Datasets Code
93      * TEMPLATE SOURCE: (None Specified.)
94      *=====*/
95      data WORK.RAW_DM ;
96      %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
97      infile '/home/u64519491/Project/raw_dm.csv' delimiter = ',' MISSOVER DSD lrecl=32767 firstobs=2 ;
98      informat STUDYID $8. ;
99      informat DOMAIN $2. ;
100     informat USUBJID $17. ;
101     informat SUBJID $8. ;
102     informat RFSTDTC yymmdd10. ;
103     informat RFENDTC yymmdd10. ;
104     informat RFXSTDTC yymmdd10. ;
105     informat RFXENDTC yymmdd10. ;
106     informat RFICDTC yymmdd10. ;
107     informat RFPENDTC yymmdd10. ;
108     informat DTHDTC $1. ;

```

```

109      informat DTHFL $1. ;
110      informat SITEID best32. ;
111      informat AGE best32. ;
112      informat AGEU $5. ;
113      informat SEX $1. ;
114      informat RACE $32. ;
115      informat ETHNIC $22. ;
116      informat ARMCD $4. ;
117      informat ARM $11. ;
118      informat ACTARMCD $4. ;
119      informat ACTARM $11. ;
120      informat COUNTRY $2. ;
121      informat DMDTC yymmdd10. ;
122      informat DMDY best32. ;
123      format STUDYID $8. ;
124      format DOMAIN $2. ;
125      format USUBJID $17. ;
126      format SUBJID $8. ;
127      format RFSTDTC yymmdd10. ;
128      format RFENDTC yymmdd10. ;
129      format RFXSTDTC yymmdd10. ;
130      format RFXENDTC yymmdd10. ;
131      format RFICDTC yymmdd10. ;
132      format RFPENDTC yymmdd10. ;
133      format DTHDTC $1. ;
134      format DTHFL $1. ;
135      format SITEID best12. ;
136      format AGE best12. ;
137      format AGEU $5. ;
138      format SEX $1. ;
139      format RACE $32. ;
140      format ETHNIC $22. ;
141      format ARMCD $4. ;
142      format ARM $11. ;
143      format ACTARMCD $4. ;
144      format ACTARM $11. ;
145      format COUNTRY $2. ;
146      format DMDTC yymmdd10. ;
147      format DMDY best12. ;
148      input
149          STUDYID $
150          DOMAIN $
151          USUBJID $
152          SUBJID $
153          RFSTDTC
154          RFENDTC
155          RFXSTDTC
156          RFXENDTC
157          RFICDTC
158          RFPENDTC
159          DTHDTC $
160          DTHFL $
161          SITEID
162          AGE
163          AGEU $
164          SEX $
165          RACE $
166          ETHNIC $
167          ARMCD $
168          ARM $
169          ACTARMCD $
170          ACTARM $
171          COUNTRY $
172          DMDTC
173          DMDY
174      ;
175      if _ERROR_ then call symputx('_EFIERR_',1); /* set ERROR detection macro variable */
176      run;

```

NOTE: The infile '/home/u64519491/Project/raw_dm.csv' is:

```

Filename=/home/u64519491/Project/raw_dm.csv,
Owner Name=u64519491,Group Name=oda,
Access Permission=-rw-r--r--,
Last Modified=25 May 2026 01:45:31,
File Size (bytes)=21102

```

NOTE: 100 records were read from the infile '/home/u64519491/Project/raw_dm.csv'.

```

The minimum record length was 195.
The maximum record length was 227.

```

NOTE: The data set WORK.RAW_DM has 100 observations and 25 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            9432.21k
OS Memory         34080.00k
Timestamp         25/05/2026 12:14:21 PM
Step Count                312  Switch Count   2
Page Faults                0
Page Reclaims             82
Page Swaps                0
Voluntary Context Switches 14
Involuntary Context Switches 0
Block Input Operations     0
Block Output Operations    272

```

100 rows created in WORK.RAW_DM from /home/u64519491/Project/raw_dm.csv.

```
NOTE: WORK.RAW_DM data set was successfully created.
NOTE: The data set WORK.RAW_DM has 100 observations and 25 variables.
NOTE: PROCEDURE IMPORT used (Total process time):
  real time           0.04 seconds
  user cpu time       0.04 seconds
  system cpu time     0.01 seconds
  memory             9432.21k
  OS Memory          34340.00k
  Timestamp           25/05/2026 12:14:21 PM
  Step Count          312  Switch Count  10
  Page Faults         0
  Page Reclaims       797
  Page Swaps          0
  Voluntary Context Switches  91
  Involuntary Context Switches  2
  Block Input Operations  0
  Block Output Operations 328
```

```
177
178      /*2. Audit the structural metadata of the imported dataset */
179      proc contents data=work.raw_dm;
180      run;
```

```
NOTE: PROCEDURE CONTENTS used (Total process time):
  real time           0.03 seconds
  user cpu time       0.04 seconds
  system cpu time     0.00 seconds
  memory             2005.96k
  OS Memory          30384.00k
  Timestamp           25/05/2026 12:14:21 PM
  Step Count          313  Switch Count  0
  Page Faults         0
  Page Reclaims       91
  Page Swaps          0
  Voluntary Context Switches  3
  Involuntary Context Switches  2
  Block Input Operations  0
  Block Output Operations 32
```

```
181
182      /*3. map it to strict cdisc SDTM Demographics standard */
183      data WORK.sdtm;
184      set work.raw_dm (RENAME=(RFSTDTc=raw_rfstdtc RFENDTC=raw_rfendtc RFXSTDTC=raw_rfxstdtc));
185      /*A. establish structural armour (length/label)*/
186      attrib
187      STUDYID LENGTH=$20 label="Study identifier"
WARNING: Length of character variable STUDYID has already been set.
      Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
188      DOMAIN LENGTH=$2 label="Domain Abbreviation"
189      USUBJID LENGTH=$40 label="Unique Subject Identifier"
WARNING: Length of character variable USUBJID has already been set.
      Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
190      SUBJID LENGTH=$10 label="Subject Identifier for Study"
WARNING: Length of character variable SUBJID has already been set.
      Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
191      RFSTDTCLength=$10 label="subject reference start date time"
192      RFENDTCLength=$10 label="subject reference end date time"
193      RFXSTDTC LENGTH=$10 LABEL="date/time of First Study Treatment"
194      AGElabel="Age"
195      AGEULENGTH=$10 label="Age Units"
WARNING: Length of character variable AGEU has already been set.
      Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
196      SEX LENGTH=$1 Label="Sex"
197      RACELENGTH=$60 Label="Race"
WARNING: Length of character variable RACE has already been set.
      Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
198      ETHNICLENGTH=$60 Label="Ethnic Group"
WARNING: Length of character variable ETHNIC has already been set.
      Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
199      ARM LENGTH=$60 Label="Description of Planned Arm";
WARNING: Length of character variable ARM has already been set.
      Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
200      /*B. execute data-type conversion (numeric to ISO8601 Character strings) */
201      RFSTDTc = PUT(raw_rfstdtc, YMMDD10.) ;
202      RFENDTC = PUT(raw_rfendtc, YMMDD10.) ;
203      RFXSTDTC = PUT(raw_rfxstdtc, YMMDD10.);
204      /*C. clean up workspace by dropping temporary numeric variables */
205      drop raw_rfstdtc raw_rfendtc raw_rfxstdtc ;
206      /*D. Audit Final Standardized Dataset structure */
```

```
NOTE: There were 100 observations read from the data set WORK.RAW_DM.
NOTE: The data set WORK.SDTM has 100 observations and 25 variables.
NOTE: DATA statement used (Total process time):
  real time           0.00 seconds
  user cpu time       0.00 seconds
  system cpu time     0.00 seconds
  memory             1132.96k
  OS Memory          30384.00k
  Timestamp           25/05/2026 12:14:21 PM
  Step Count          314  Switch Count  2
  Page Faults         0
  Page Reclaims       115
  Page Swaps          0
```

```

Voluntary Context Switches      14
Involuntary Context Switches    1
Block Input Operations           0
Block Output Operations         264

```

```

207      proc contents data=work.sdtm;
208      run;

```

```

NOTE: PROCEDURE CONTENTS used (Total process time):
real time          0.04 seconds
user cpu time      0.04 seconds
system cpu time    0.00 seconds
memory            938.40k
OS Memory         30384.00k
Timestamp         25/05/2026 12:14:21 PM
Step Count        315  Switch Count  0
Page Faults       0
Page Reclaims     91
Page Swaps        0
Voluntary Context Switches    3
Involuntary Context Switches  1
Block Input Operations        0
Block Output Operations      32

```

```

209
210      /*4. Audit Categorical Values for Controlled Terminology Compliance */
211      proc freq data=work.sdtm;
212      tables SEX RACE ETHNIC ARM ARMCD ;
213      run;

```

NOTE: There were 100 observations read from the data set WORK.SDTM.

```

NOTE: PROCEDURE FREQ used (Total process time):
real time          0.03 seconds
user cpu time      0.03 seconds
system cpu time    0.00 seconds
memory            1087.78k
OS Memory         30384.00k
Timestamp         25/05/2026 12:14:21 PM
Step Count        316  Switch Count  2
Page Faults       0
Page Reclaims    119
Page Swaps        0
Voluntary Context Switches    15
Involuntary Context Switches  2
Block Input Operations        0
Block Output Operations      280

```

```

214
215      /*5. Lock the Clean Dataset into Final Output Workspace */
216      data work.dm_final;
217      set work.sdtm;
218      SEX = upcase(strip(SEX));
219      ARMCD = upcase(strip(ARMCD));
220      IF MISSING(USUBJID) then PUT "WARNING: Missing USUBJID for Observation"_N_;

```

49

NOTE 49-169: The meaning of an identifier after a quoted string might change in a future SAS release. Inserting white space between a quoted string and the succeeding identifier is recommended.

```

221      run;

```

NOTE: There were 100 observations read from the data set WORK.SDTM.
NOTE: The data set WORK.DM_FINAL has 100 observations and 25 variables.

```

NOTE: DATA statement used (Total process time):
real time          0.00 seconds
user cpu time      0.01 seconds
system cpu time    0.00 seconds
memory            1135.31k
OS Memory         30384.00k
Timestamp         25/05/2026 12:14:21 PM
Step Count        317  Switch Count  2
Page Faults       0
Page Reclaims    115
Page Swaps        0
Voluntary Context Switches    18
Involuntary Context Switches  0
Block Input Operations        0
Block Output Operations      264

```

```

222
223      /*=====
224      * PHASE 2: THE ADVERSE EVENTS OBSERVED DOMAIN (AE)- Working on ADVERSE EVENT File from CDISC-DATASET-GENERATOR
225      *=====
226
227      /*1. Ingest Raw Adverse Events Data */
228      proc import datafile="/home/u64519491/Project/raw_ae.csv"
229      out=work.raw_ae dbms=csv replace;
229      !                               getnames=YES;
230      run;

```

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to WORK.PARMS.PARMS.SLIST.

```

231      /*****
232      *   PRODUCT:   SAS

```

```

233 *   VERSION:   9.4
234 *   CREATOR:   External File Interface
235 *   DATE:      25MAY26
236 *   DESC:      Generated SAS Datasets Code
237 *   TEMPLATE SOURCE: (None Specified.)
238 *****/
239 data WORK.RAW_AE ;
240 %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
241 infile '/home/u64519491/Project/raw_ae.csv' delimiter = ',' MISOVER DSD lrecl=32767 firstobs=2 ;
242 informat STUDYID $8. ;
243 informat DOMAIN $2. ;
244 informat USUBJID $17. ;
245 informat AESEQ best32. ;
246 informat AESPID $3. ;
247 informat AETERM $9. ;
248 informat AEDECOD $9. ;
249 informat AEBODSYS $26. ;
250 informat AESOC $26. ;
251 informat AESEV $8. ;
252 informat AETOXGR best32. ;
253 informat AESER $1. ;
254 informat AEREL $11. ;
255 informat AEACN $16. ;
256 informat AEOUT $26. ;
257 informat AESTDTC yymmdd10. ;
258 informat AEENDTC yymmdd10. ;
259 informat AESTDY best32. ;
260 informat AEENDY best32. ;
261 informat AEDUR best32. ;
262 informat AECONTRT $1. ;
263 informat AESDTH $1. ;
264 informat AESLIFE $1. ;
265 informat AESHOSP $1. ;
266 informat AESDISAB $1. ;
267 informat AESMIE $1. ;
268 informat AEACNOTH $1. ;
269 format STUDYID $8. ;
270 format DOMAIN $2. ;
271 format USUBJID $17. ;
272 format AESEQ best12. ;
273 format AESPID $3. ;
274 format AETERM $9. ;
275 format AEDECOD $9. ;
276 format AEBODSYS $26. ;
277 format AESOC $26. ;
278 format AESEV $8. ;
279 format AETOXGR best12. ;
280 format AESER $1. ;
281 format AEREL $11. ;
282 format AEACN $16. ;
283 format AEOUT $26. ;
284 format AESTDTC yymmdd10. ;
285 format AEENDTC yymmdd10. ;
286 format AESTDY best12. ;
287 format AEENDY best12. ;
288 format AEDUR best12. ;
289 format AECONTRT $1. ;
290 format AESDTH $1. ;
291 format AESLIFE $1. ;
292 format AESHOSP $1. ;
293 format AESDISAB $1. ;
294 format AESMIE $1. ;
295 format AEACNOTH $1. ;
296 input
297     STUDYID $
298     DOMAIN $
299     USUBJID $
300     AESEQ
301     AESPID $
302     AETERM $
303     AEDECOD $
304     AEBODSYS $
305     AESOC $
306     AESEV $
307     AETOXGR
308     AESER $
309     AEREL $
310     AEACN $
311     AEOUT $
312     AESTDTC
313     AEENDTC
314     AESTDY
315     AEENDY
316     AEDUR
317     AECONTRT $
318     AESDTH $
319     AESLIFE $
320     AESHOSP $
321     AESDISAB $
322     AESMIE $
323     AEACNOTH $
324 ;
325 if _ERROR_ then call symputx('_EFIERR_',1); /* set ERROR detection macro variable */
326 run;

```

NOTE: The infile '/home/u64519491/Project/raw_ae.csv' is:
 Filename=/home/u64519491/Project/raw_ae.csv,
 Owner Name=u64519491, Group Name=oda,


```
Access Permission=-rw-r--r--,
Last Modified=25 May 2026 01:59:09,
File Size (bytes)=49362
```

NOTE: 254 records were read from the infile '/home/u64519491/Project/raw_ae.csv'.

```
The minimum record length was 154.
The maximum record length was 220.
```

NOTE: The data set WORK.RAW_AE has 254 observations and 27 variables.

NOTE: DATA statement used (Total process time):

```
real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            9324.40k
OS Memory         34080.00k
Timestamp         25/05/2026 12:14:21 PM
Step Count        318  Switch Count  2
Page Faults       0
Page Reclaims     73
Page Swaps        0
Voluntary Context Switches 14
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 280
```

254 rows created in WORK.RAW_AE from /home/u64519491/Project/raw_ae.csv.

NOTE: WORK.RAW_AE data set was successfully created.

NOTE: The data set WORK.RAW_AE has 254 observations and 27 variables.

NOTE: PROCEDURE IMPORT used (Total process time):

```
real time          0.04 seconds
user cpu time      0.03 seconds
system cpu time    0.01 seconds
memory            9324.40k
OS Memory         34340.00k
Timestamp         25/05/2026 12:14:21 PM
Step Count        318  Switch Count 10
Page Faults       0
Page Reclaims    786
Page Swaps        0
Voluntary Context Switches 107
Involuntary Context Switches 3
Block Input Operations 0
Block Output Operations 296
```

```
327
328      /*2. Sort AE data */
329      proc sort data=work.raw_ae out=work.ae_sorted;
330      by USUBJID AESTDTC AETERM ;
331      run;
```

NOTE: There were 254 observations read from the data set WORK.RAW_AE.

NOTE: The data set WORK.AE_SORTED has 254 observations and 27 variables.

NOTE: PROCEDURE SORT used (Total process time):

```
real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.01 seconds
memory            1314.56k
OS Memory         30644.00k
Timestamp         25/05/2026 12:14:21 PM
Step Count        319  Switch Count  2
Page Faults       0
Page Reclaims    132
Page Swaps        0
Voluntary Context Switches 20
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 272
```

```
332
333      /*3. Map/Structure/Generate SDTM AE Architecture */
334      data work.ae_final;
335      set work.ae_sorted (rename=(AESTDTC=raw_aestdct AEENDTC=raw_aeendtc));
336      by USUBJID;
337      attrib
338      STUDYID length=$20 label="Study Identifier"
```

WARNING: Length of character variable STUDYID has already been set.
Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.

```
339      DOMAIN length=$2 label="Domain Abbreviation"
340      USUBJID length=$40 label="Unique Subject Identifier"
```

WARNING: Length of character variable USUBJID has already been set.
Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.

```
341      AESEQ label="Adverse Event Sequence Number"
342      AETERM length=$200 label="Reported Term for the Adverse Event"
```

WARNING: Length of character variable AETERM has already been set.
Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.

```
343      AEDECOD length=$200 label="Dictionary-Derived Term"
```

WARNING: Length of character variable AEDECOD has already been set.
Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.

```
344      AESTDTC length=$10 label="Start Date/Time of Adverse Event"
```

```
345      AEENDTC length=$10 label="End Date/Time of Adverse Event"
```

```
346      AESEV length=$5 label="Severity"
```

WARNING: Length of character variable AESEV has already been set.
Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.


```

347      AESER length=$1 label="Serious Event" ;
348      /* A. set domain code B. formating Dates and Terms
349      C. control terminology D. Logic on Sequence Number*/
350      DOMAIN = "AE" ;
351      AEDECOD = upcase(strip(AETERM));
352      if raw_aestdtc ne . then AESTDTC = put(raw_aestdtc, yymmdd10.) ;
353      if raw_aeendtc ne . then AEENDTC = put(raw_aeendtc, yymmdd10.) ;
354      AESEV = upcase(strip(AESEV));
355      AESER = upcase(strip(AESER));
356      retain AESEQ;
357      if first.USUBJID then AESEQ = 1 ;
358      else AESEQ = AESEQ+1 ;
359      drop raw_aestdtc raw_aeendtc;
360      run;

```

NOTE: There were 254 observations read from the data set WORK.AE_SORTED.

NOTE: The data set WORK.AE_FINAL has 254 observations and 27 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            1190.15k
OS Memory          30384.00k
Timestamp          25/05/2026 12:14:21 PM
Step Count         320  Switch Count  2
Page Faults        0
Page Reclaims      106
Page Swaps         0
Voluntary Context Switches  12
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations  272

```

```

361
362      /*===== *
363      PHASE 3: REGULATORY QUALITY CONTROL & INTEGRITY AUDIT
364      * =====*/
365
366      /*1. verify Meta-Data Constraints for DM Domain */
367      proc contents data=work.dm_final;
368      run;

```

NOTE: PROCEDURE CONTENTS used (Total process time):

```

real time          0.04 seconds
user cpu time      0.04 seconds
system cpu time    0.00 seconds
memory            943.43k
OS Memory          30384.00k
Timestamp          25/05/2026 12:14:21 PM
Step Count         321  Switch Count  0
Page Faults        0
Page Reclaims      91
Page Swaps         0
Voluntary Context Switches  3
Involuntary Context Switches 2
Block Input Operations  0
Block Output Operations  24

```

```

369
370      /*2. Verify Meta-Data Constraints for AE Domain */
371      proc contents data=work.ae_final;
372      run;

```

NOTE: PROCEDURE CONTENTS used (Total process time):

```

real time          0.04 seconds
user cpu time      0.04 seconds
system cpu time    0.00 seconds
memory            1056.56k
OS Memory          30384.00k
Timestamp          25/05/2026 12:14:21 PM
Step Count         322  Switch Count  0
Page Faults        0
Page Reclaims      91
Page Swaps         0
Voluntary Context Switches  3
Involuntary Context Switches 2
Block Input Operations  0
Block Output Operations  24

```

```

373
374      /*3. Cross-Domain Validation: Confirm all AE patients exist in DM master */
375      proc sql;
376      select count(distinct USUBJID) as Orphaned_AE_Patients
377      from work.ae_final
378      where USUBJID not in (select distinct USUBJID from work.dm_final);
379      quit;

```

NOTE: PROCEDURE SQL used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            5887.81k
OS Memory          35508.00k
Timestamp          25/05/2026 12:14:21 PM
Step Count         323  Switch Count  0
Page Faults        0

```

```

Page Reclaims          89
Page Swaps              0
Voluntary Context Switches  3
Involuntary Context Switches 0
Block Input Operations    0
Block Output Operations   40

```

```

380
381      /*4. Display Final Standardized Data Samples */
382      proc print data=work.dm_final (obs=5);
383      title " SDTM Demographics (DM) Final Compliance View " ;
384      run;

```

NOTE: There were 5 observations read from the data set WORK.DM_FINAL.

NOTE: PROCEDURE PRINT used (Total process time):

```

real time          0.02 seconds
user cpu time      0.03 seconds
system cpu time    0.00 seconds
memory            1032.96k
OS Memory          30124.00k
Timestamp          25/05/2026 12:14:22 PM
Step Count         324  Switch Count  1
Page Faults        0
Page Reclaims      61
Page Swaps         0
Voluntary Context Switches 10
Involuntary Context Switches 3
Block Input Operations 0
Block Output Operations 0

```

```

385      proc print data=work.ae_final (obs=5);
386      title " SDTM Adverse Events (AE) Final Compliance View " ;
387      run;

```

NOTE: There were 5 observations read from the data set WORK.AE_FINAL.

NOTE: PROCEDURE PRINT used (Total process time):

```

real time          0.02 seconds
user cpu time      0.02 seconds
system cpu time    0.00 seconds
memory            818.31k
OS Memory          30124.00k
Timestamp          25/05/2026 12:14:22 PM
Step Count         325  Switch Count  1
Page Faults        0
Page Reclaims      61
Page Swaps         0
Voluntary Context Switches 9
Involuntary Context Switches 3
Block Input Operations 0
Block Output Operations 16

```

```

388      title;
389
390      /*=====
391      * PHASE 4: THE SQL RELATIONAL JOIN & SAFETY PROFILE TFL
392      *=====*/
393
394      /* 1. Execute relational join and standardize text directly in the database */
395      proc sql;
396      create table work.safety_profile as
397      select
398      dm.USUBJID,
399      dm.AGE,
400      dm.SEX,
401      dm.RACE,
402      dm.ARM,
403      ae.AESEQ,
404      ae.AETERM,
405      ae.AEDECOD,
406      case
407      when missing(ae.AESEV) then "No Adverse Event"
408      else propcase(strip(ae.AESEV))
409      end as AESEV length=16 label="Severity/Intensity",
410      ae.AEREL
411      from work.dm_final as dm
412      left join work.ae_final as ae
413      on dm.USUBJID = ae.USUBJID;

```

NOTE: Table WORK.SAFETY_PROFILE created, with 266 rows and 10 columns.

```

414      quit;
NOTE: PROCEDURE SQL used (Total process time):
real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            5995.00k
OS Memory          35508.00k
Timestamp          25/05/2026 12:14:22 PM
Step Count         326  Switch Count  2
Page Faults        0
Page Reclaims      216
Page Swaps         0
Voluntary Context Switches 11
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 280

```

```

415
416      /* 2. Generate the final, publication-ready Clinical Safety Summary Report */
417      proc freq data=work.safety_profile;
418      tables ARM * AESEV / nocol nopercnt norow;
419      title "Safety Audit: Comprehensive Population Coverage (Including No-AE Subjects)";
420      run;

```

NOTE: There were 266 observations read from the data set WORK.SAFETY_PROFILE.

NOTE: PROCEDURE FREQ used (Total process time):

```

real time          0.01 seconds
user cpu time      0.02 seconds
system cpu time    0.00 seconds
memory            1183.53k
OS Memory          30644.00k
Timestamp          25/05/2026 12:14:22 PM
Step Count         327  Switch Count  5
Page Faults        0
Page Reclaims      184
Page Swaps         0
Voluntary Context Switches  36
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations 528

```

```

421      title;
422
423      /*=====
424      * PHASE 5: PRODUCTION TFL - CLINICAL DATA LISTING (NEW)
425      *=====*/
426
427      options nodate nonumber orientation=landscape;
428      ods listing close;
429
430      /* Generate Clinical-Grade Subject Adverse Event Listing */
431      proc report data=work.safety_profile nowd headskip;
432      where AESEV ne "No Adverse Event";
433      column ARM USUBJID AESEQ AEDECOD AESEV AEREL;
434
435      define ARM      / order "Treatment Arm" width=15;
436      define USUBJID  / order "Unique Subject ID" width=25;
437      define AESEQ    / display "AE Seq" center width=8;
438      define AEDECOD  / display "Standardized Term (MedDRA)" width=30;
439      define AESEV    / display "Severity" width=15;
440      define AEREL    / display "Causality" width=15;
441
442      break after ARM / skip;
443
444      title1 "Protocol: STUDY001";
445      title2 "Listing 16.2.7: Adverse Events Log - Patients with Documented Events";
446      run;

```

NOTE: There were 254 observations read from the data set WORK.SAFETY_PROFILE.

WHERE AESEV not = 'No Adverse Event';

NOTE: PROCEDURE REPORT used (Total process time):

```

real time          0.16 seconds
user cpu time      0.16 seconds
system cpu time    0.00 seconds
memory            7508.93k
OS Memory          35264.00k
Timestamp          25/05/2026 12:14:22 PM
Step Count         328  Switch Count  5
Page Faults        0
Page Reclaims      1479
Page Swaps         0
Voluntary Context Switches  44
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations 112

```

```

447      title;
448      options date number orientation=portrait;
449
450      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
460

```

Results: Project.sas

The CONTENTS Procedure

Data Set Name	WORK.RAW_DM	Observations	100
Member Type	DATA	Variables	25
Engine	V9	Indexes	0
Created	25/05/2026 17:44:22	Observation Length	216
Last Modified	25/05/2026 17:44:22	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	606
Obs in First Data Page	100
Number of Data Set Repairs	0
Filename	/saswork/SAS_work5BFE00017EA0_odaws01-apse1.oda.sas.com/SAS_workBF4100017EA0_odaws01-apse1.oda.sas.com/raw_dm.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1074793174
Access Permission	rw-r--r--
Owner Name	u64519491
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
22	ACTARM	Char	11	\$11.	\$11.
21	ACTARMCD	Char	4	\$4.	\$4.
14	AGE	Num	8	BEST12.	BEST32.
15	AGEU	Char	5	\$5.	\$5.
20	ARM	Char	11	\$11.	\$11.
19	ARMCD	Char	4	\$4.	\$4.
23	COUNTRY	Char	2	\$2.	\$2.
24	DMDTC	Num	8	YYMMDD10.	YYMMDD10.
25	DMDY	Num	8	BEST12.	BEST32.
2	DOMAIN	Char	2	\$2.	\$2.
11	DTHDTC	Char	1	\$1.	\$1.
12	DTHFL	Char	1	\$1.	\$1.
18	ETHNIC	Char	22	\$22.	\$22.
17	RACE	Char	32	\$32.	\$32.
6	RFENDTC	Num	8	YYMMDD10.	YYMMDD10.
9	RFICDTC	Num	8	YYMMDD10.	YYMMDD10.
10	RFPENDTC	Num	8	YYMMDD10.	YYMMDD10.
5	RFSTDTC	Num	8	YYMMDD10.	YYMMDD10.
8	RFXENDTC	Num	8	YYMMDD10.	YYMMDD10.
7	RFXSTDTC	Num	8	YYMMDD10.	YYMMDD10.
16	SEX	Char	1	\$1.	\$1.
13	SITEID	Num	8	BEST12.	BEST32.
1	STUDYID	Char	8	\$8.	\$8.
4	SUBJID	Char	8	\$8.	\$8.
3	USUBJID	Char	17	\$17.	\$17.

The CONTENTS Procedure

Data Set Name	WORK.SDTM	Observations	100
Member Type	DATA	Variables	25
Engine	V9	Indexes	0
Created	25/05/2026 17:44:22	Observation Length	216
Last Modified	25/05/2026 17:44:22	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	606
Obs in First Data Page	100
Number of Data Set Repairs	0
Filename	/saswork/SAS_work5BFE00017EA0_odaws01-apse1.oda.sas.com/SAS_workBF4100017EA0_odaws01-apse1.oda.sas.com/sdtm.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1074793168
Access Permission	rw-r--r--
Owner Name	u64519491
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
19	ACTARM	Char	11	\$11.	\$11.	
18	ACTARMCD	Char	4	\$4.	\$4.	
11	AGE	Num	8	BEST12.	BEST32.	Age
12	AGEU	Char	5	\$5.	\$5.	Age Units
17	ARM	Char	11	\$11.	\$11.	Description of Planned Arm
16	ARMCD	Char	4	\$4.	\$4.	
20	COUNTRY	Char	2	\$2.	\$2.	
21	DMDTC	Num	8	YYMMDD10.	YYMMDD10.	
22	DMDY	Num	8	BEST12.	BEST32.	

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
2	DOMAIN	Char	2	\$2.	\$2.	Domain Abbreviation
8	DTHDTC	Char	1	\$1.	\$1.	
9	DTHFL	Char	1	\$1.	\$1.	
15	ETHNIC	Char	22	\$22.	\$22.	Ethnic Group
14	RACE	Char	32	\$32.	\$32.	Race
24	RFENDTC	Char	10			subject reference end date time
6	RFICDTC	Num	8	YYMMDD10.	YYMMDD10.	
7	RFPENDTC	Num	8	YYMMDD10.	YYMMDD10.	
23	RFSTDTC	Char	10			subject reference start date time
5	RFXENDTC	Num	8	YYMMDD10.	YYMMDD10.	
25	RFXSTDTC	Char	10			date/time of First Study Treatment
13	SEX	Char	1	\$1.	\$1.	Sex
10	SITEID	Num	8	BEST12.	BEST32.	
1	STUDYID	Char	8	\$8.	\$8.	Study identifier
4	SUBJID	Char	8	\$8.	\$8.	Subject Identifier for Study
3	USUBJID	Char	17	\$17.	\$17.	Unique Subject Identifier

The FREQ Procedure

Sex				
SEX	Frequency	Percent	Cumulative Frequency	Cumulative Percent
F	42	42.00	42	42.00
M	58	58.00	100	100.00

Race				
RACE	Frequency	Percent	Cumulative Frequency	Cumulative Percent
AMERICAN INDIAN OR ALASKA NATIVE	27	27.00	27	27.00
ASIAN	19	19.00	46	46.00
BLACK OR AFRICAN AMERICAN	16	16.00	62	62.00
OTHER	21	21.00	83	83.00
WHITE	17	17.00	100	100.00

Ethnic Group				
ETHNIC	Frequency	Percent	Cumulative Frequency	Cumulative Percent
HISPANIC OR LATINO	44	44.00	44	44.00
NOT HISPANIC OR LATINO	56	56.00	100	100.00

Description of Planned Arm				
ARM	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Treatment 1	50	50.00	50	50.00
Treatment 2	50	50.00	100	100.00

ARMCD	Frequency	Percent	Cumulative Frequency	Cumulative Percent
ARM1	50	50.00	50	50.00
ARM2	50	50.00	100	100.00

The CONTENTS Procedure

Data Set Name	WORK.DM_FINAL	Observations	100
Member Type	DATA	Variables	25
Engine	V9	Indexes	0
Created	25/05/2026 17:44:22	Observation Length	216
Last Modified	25/05/2026 17:44:22	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	606
Obs in First Data Page	100
Number of Data Set Repairs	0
Filename	/saswork/SAS_work5BFE00017EA0_odaws01-apse1.oda.sas.com/SAS_workBF4100017EA0_odaws01-apse1.oda.sas.com/dm_final.sas7bdatt
Release Created	9.0401M8
Host Created	Linux
Inode Number	1074793180
Access Permission	rw-r--r--
Owner Name	u64519491
File Size	256KB

Engine/Host Dependent Information	
File Size (bytes)	262144

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
19	ACTARM	Char	11	\$11.	\$11.	
18	ACTARMCD	Char	4	\$4.	\$4.	
11	AGE	Num	8	BEST12.	BEST32.	Age
12	AGEU	Char	5	\$5.	\$5.	Age Units
17	ARM	Char	11	\$11.	\$11.	Description of Planned Arm
16	ARMCD	Char	4	\$4.	\$4.	
20	COUNTRY	Char	2	\$2.	\$2.	
21	DMDTC	Num	8	YYMMDD10.	YYMMDD10.	
22	DMDY	Num	8	BEST12.	BEST32.	
2	DOMAIN	Char	2	\$2.	\$2.	Domain Abbreviation
8	DTHDTC	Char	1	\$1.	\$1.	
9	DTHFL	Char	1	\$1.	\$1.	
15	ETHNIC	Char	22	\$22.	\$22.	Ethnic Group
14	RACE	Char	32	\$32.	\$32.	Race
24	RFENDTC	Char	10			subject reference end date time
6	RFICDTC	Num	8	YYMMDD10.	YYMMDD10.	
7	RFPENDTC	Num	8	YYMMDD10.	YYMMDD10.	
23	RFSTDTC	Char	10			subject reference start date time
5	RFXENDTC	Num	8	YYMMDD10.	YYMMDD10.	
25	RFXSTDTC	Char	10			date/time of First Study Treatment
13	SEX	Char	1	\$1.	\$1.	Sex
10	SITEID	Num	8	BEST12.	BEST32.	
1	STUDYID	Char	8	\$8.	\$8.	Study identifier
4	SUBJID	Char	8	\$8.	\$8.	Subject Identifier for Study
3	USUBJID	Char	17	\$17.	\$17.	Unique Subject Identifier

The CONTENTS Procedure

Data Set Name	WORK.AE_FINAL	Observations	254
Member Type	DATA	Variables	27
Engine	V9	Indexes	0
Created	25/05/2026 17:44:22	Observation Length	232
Last Modified	25/05/2026 17:44:22	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	564
Obs in First Data Page	254
Number of Data Set Repairs	0
Filename	/saswork/SAS_work5BFE00017EA0_odaws01-apse1.oda.sas.com/SAS_workBF4100017EA0_odaws01-apse1.oda.sas.com/ae_final.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1074793162
Access Permission	rw-r--r--
Owner Name	u64519491
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
14	AEACN	Char	16	\$16.	\$16.	
25	AEACNOTH	Char	1	\$1.	\$1.	
8	AEBODSYS	Char	26	\$26.	\$26.	
19	AECONTRT	Char	1	\$1.	\$1.	
7	AEECOD	Char	9	\$9.	\$9.	Dictionary-Derived Term
18	AEDUR	Num	8	BEST12.	BEST32.	
27	AEENDTC	Char	10			End Date/Time of Adverse Event
17	AEENDY	Num	8	BEST12.	BEST32.	
15	AEOUT	Char	26	\$26.	\$26.	
13	AEREL	Char	11	\$11.	\$11.	
23	AESDISAB	Char	1	\$1.	\$1.	
20	AESDTH	Char	1	\$1.	\$1.	
4	AESQ	Num	8	BEST12.	BEST32.	Adverse Event Sequence Number
12	AESER	Char	1	\$1.	\$1.	Serious Event
10	AESV	Char	8	\$8.	\$8.	Severity
22	AESHOSP	Char	1	\$1.	\$1.	
21	AESLIFE	Char	1	\$1.	\$1.	
24	AESMIE	Char	1	\$1.	\$1.	
9	AESOC	Char	26	\$26.	\$26.	
5	AESPID	Char	3	\$3.	\$3.	
26	AESTDTC	Char	10			Start Date/Time of Adverse Event
16	AESTDY	Num	8	BEST12.	BEST32.	

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
6	AETERM	Char	9	\$9.	\$9.	Reported Term for the Adverse Event
11	AETOXGR	Num	8	BEST12.	BEST32.	
2	DOMAIN	Char	2	\$2.	\$2.	Domain Abbreviation
1	STUDYID	Char	8	\$8.	\$8.	Study Identifier
3	USUBJID	Char	17	\$17.	\$17.	Unique Subject Identifier

Orphaned_AE_Patients
0

SDTM Demographics (DM) Final Compliance View

Obs	STUDYID	DOMAIN	USUBJID	SUBJID	RFXENDTC	RFICDTC	RFPENDTC	DTHDTC	DTHFL	SITEID	AGE	AGEU	SEX	RACE	ETHNIC	ARMCD	ARM	ACTARMCD	ACT/
1	STUDY001	DM	STUDY001-SUBJ0001	SUBJ0001	2026-05-14	2026-02-22	2026-05-13			44	45	YEARS	M	WHITE	HISPANIC OR LATINO	ARM2	Treatment 2	ARM2	Treat 2
2	STUDY001	DM	STUDY001-SUBJ0002	SUBJ0002	2026-05-17	2026-01-26	2026-05-10			46	76	YEARS	M	BLACK OR AFRICAN AMERICAN	NOT HISPANIC OR LATINO	ARM2	Treatment 2	ARM1	Treat 1
3	STUDY001	DM	STUDY001-SUBJ0003	SUBJ0003	2026-04-30	2026-02-04	2026-05-07			32	62	YEARS	F	BLACK OR AFRICAN AMERICAN	NOT HISPANIC OR LATINO	ARM2	Treatment 2	ARM2	Treat 2
4	STUDY001	DM	STUDY001-SUBJ0004	SUBJ0004	2026-05-06	2026-01-18	2026-04-29			42	56	YEARS	F	AMERICAN INDIAN OR ALASKA NATIVE	HISPANIC OR LATINO	ARM1	Treatment 1	ARM1	Treat 1
5	STUDY001	DM	STUDY001-SUBJ0005	SUBJ0005	2026-05-15	2025-06-21	2026-05-14			18	60	YEARS	F	ASIAN	HISPANIC OR LATINO	ARM2	Treatment 2	ARM2	Treat 2

SDTM Adverse Events (AE) Final Compliance View

Obs	STUDYID	DOMAIN	USUBJID	AESQ	AESPID	AETERM	AEDECOD	AEBODSYS	AESOC	AESQ	AETOXGR	AESER	AEREL	AEACN	AEOUT
1	STUDY001	AE	STUDY001-SUBJ0001	1	AE1	NAUSEA	NAUSEA	GASTROINTESTINAL DISORDERS	GASTROINTESTINAL DISORDERS	SEVERE	3	Y	NOT RELATED	DRUG WITHDRAWN	RECOVERE
2	STUDY001	AE	STUDY001-SUBJ0001	4	AE3	VOMITING	VOMITING	NERVOUS SYSTEM DISORDERS	NERVOUS SYSTEM DISORDERS	MODERATE	2	N	NOT RELATED	DRUG WITHDRAWN	NOT RECOV RESOLVED
3	STUDY001	AE	STUDY001-SUBJ0001	5	AE4	HEADACHE	HEADACHE	CARDIAC DISORDERS	CARDIAC DISORDERS	SEVERE	3	Y	RELATED	DRUG INTERRUPTED	RECOVERIN
4	STUDY001	AE	STUDY001-SUBJ0001	6	AE5	FATIGUE	FATIGUE	RESPIRATORY DISORDERS	RESPIRATORY DISORDERS	MILD	1	N	NOT RELATED	NOT APPLICABLE	FATAL
5	STUDY001	AE	STUDY001-SUBJ0001	3	AE2	FEVER	FEVER	RESPIRATORY DISORDERS	RESPIRATORY DISORDERS	MODERATE	2	N	RELATED	NOT APPLICABLE	RECOVERIN

Safety Audit: Comprehensive Population Coverage (Including No-AE Subjects)

The FREQ Procedure

Frequency	Table of ARM by AESEV					
	AESEV(Severity/Intensity)					Total
	ARM(Description of Planned Arm)	Mild	Moderate	No Adverse Event	Severe	
	Treatment 1	43	40	7	37	127
	Treatment 2	46	45	5	43	139
	Total	89	85	12	80	266

Protocol: STUDY001
Listing 16.2.7: Adverse Events Log - Patients with Documented Events

Treatment Arm	Unique Subject ID	AE Seq	Standardized Term (MedDRA)	Severity	Causality
Treatment 1	STUDY001-SUBJ0004	1	FEVER	Severe	RELATED
		1	FATIGUE	Mild	RELATED
		2	DIZZINESS	Moderate	NOT RELATED
		3	HEADACHE	Moderate	RELATED
	STUDY001-SUBJ0007	1	COUGH	Mild	RELATED
		2	HEADACHE	Severe	RELATED
	STUDY001-SUBJ0009	3	DIARRHEA	Severe	RELATED
		4	FEVER	Moderate	NOT RELATED
		1	COUGH	Severe	RELATED
	STUDY001-SUBJ0010	3	HEADACHE	Mild	NOT RELATED
		1	FATIGUE	Moderate	NOT RELATED
		5	FEVER	Mild	NOT RELATED
		4	FEVER	Moderate	RELATED
	STUDY001-SUBJ0012	1	DIARRHEA	Moderate	NOT RELATED
		5	FEVER	Moderate	NOT RELATED
		4	PAIN	Moderate	NOT RELATED
		3	RASH	Moderate	RELATED
	STUDY001-SUBJ0013	1	FEVER	Moderate	RELATED
	STUDY001-SUBJ0015	3	DIZZINESS	Severe	RELATED
		1	DIARRHEA	Moderate	RELATED
	STUDY001-SUBJ0017	2	NAUSEA	Moderate	NOT RELATED
		1	FATIGUE	Mild	RELATED

Treatment Arm	Unique Subject ID	AE Seq	Standardized Term (MedDRA)	Severity	Causality
	STUDY001-SUBJ0019	6	COUGH	Mild	NOT RELATED
		2	VOMITING	Severe	RELATED
		1	RASH	Severe	RELATED
		5	PAIN	Severe	RELATED
		4	NAUSEA	Severe	NOT RELATED
	STUDY001-SUBJ0024	1	RASH	Moderate	RELATED
		3	PAIN	Severe	NOT RELATED
	STUDY001-SUBJ0025	4	FATIGUE	Severe	RELATED
		1	HEADACHE	Mild	RELATED
		5	RASH	Severe	NOT RELATED
		3	COUGH	Mild	NOT RELATED
	STUDY001-SUBJ0026	4	PAIN	Severe	RELATED
		3	DIZZINESS	Mild	NOT RELATED
		6	FEVER	Severe	NOT RELATED
		2	VOMITING	Mild	RELATED
		1	RASH	Moderate	RELATED
	STUDY001-SUBJ0028	3	FEVER	Severe	RELATED
		1	INSOMNIA	Severe	NOT RELATED
	STUDY001-SUBJ0029	2	NAUSEA	Mild	NOT RELATED
		1	PAIN	Severe	RELATED
	STUDY001-SUBJ0031	2	DIZZINESS	Mild	NOT RELATED
		4	RASH	Mild	RELATED
		1	VOMITING	Severe	NOT RELATED
	STUDY001-SUBJ0032	3	DIARRHEA	Moderate	NOT RELATED
		4	FEVER	Moderate	NOT RELATED
		5	NAUSEA	Moderate	NOT RELATED
		2	COUGH	Mild	NOT RELATED
		1	VOMITING	Moderate	NOT RELATED
	STUDY001-SUBJ0034	3	FATIGUE	Mild	RELATED
		1	HEADACHE	Moderate	RELATED
	STUDY001-SUBJ0035	4	DIARRHEA	Severe	RELATED
		2	COUGH	Mild	NOT RELATED
		1	PAIN	Moderate	RELATED
	STUDY001-SUBJ0038	6	PAIN	Moderate	NOT RELATED
		2	FEVER	Moderate	RELATED
		4	NAUSEA	Mild	NOT RELATED
		5	FEVER	Moderate	RELATED
		1	NAUSEA	Severe	NOT RELATED
	STUDY001-SUBJ0044	3	HEADACHE	Moderate	NOT RELATED
		1	HEADACHE	Mild	RELATED
	STUDY001-SUBJ0046	3	VOMITING	Severe	NOT RELATED
		1	RASH	Mild	NOT RELATED
	STUDY001-SUBJ0047	2	COUGH	Severe	RELATED
		1	VOMITING	Moderate	RELATED
	STUDY001-SUBJ0049	6	DIZZINESS	Severe	RELATED
		2	COUGH	Mild	RELATED
		1	VOMITING	Mild	NOT RELATED
		5	HEADACHE	Severe	NOT RELATED
		4	COUGH	Moderate	NOT RELATED
	STUDY001-SUBJ0051	4	FEVER	Moderate	RELATED
		6	VOMITING	Severe	NOT RELATED
		2	INSOMNIA	Severe	RELATED
		5	VOMITING	Mild	NOT RELATED
		1	VOMITING	Moderate	NOT RELATED
	STUDY001-SUBJ0056	1	FATIGUE	Mild	NOT RELATED
	STUDY001-SUBJ0059	6	COUGH	Mild	RELATED
		4	PAIN	Moderate	RELATED
		2	FEVER	Severe	NOT RELATED
		5	INSOMNIA	Mild	NOT RELATED
		1	PAIN	Mild	RELATED
	STUDY001-SUBJ0063	1	INSOMNIA	Mild	RELATED
		3	HEADACHE	Severe	RELATED
		1	PAIN	Mild	NOT RELATED
	STUDY001-SUBJ0070	2	DIZZINESS	Mild	RELATED
		1	VOMITING	Moderate	RELATED
	STUDY001-SUBJ0075	3	RASH	Moderate	NOT RELATED
		1	FATIGUE	Moderate	RELATED
	STUDY001-SUBJ0077	5	HEADACHE	Moderate	RELATED
		2	PAIN	Severe	NOT RELATED
		4	HEADACHE	Mild	NOT RELATED
		3	RASH	Mild	RELATED
		1	FEVER	Severe	NOT RELATED
	STUDY001-SUBJ0079	1	HEADACHE	Moderate	NOT RELATED
	STUDY001-SUBJ0080	1	COUGH	Mild	NOT RELATED
	STUDY001-SUBJ0081	4	HEADACHE	Mild	NOT RELATED
		3	FEVER	Severe	NOT RELATED
		6	DIZZINESS	Severe	NOT RELATED
		2	INSOMNIA	Mild	RELATED
		1	NAUSEA	Mild	RELATED
	STUDY001-SUBJ0083	3	NAUSEA	Moderate	NOT RELATED
		1	COUGH	Mild	NOT RELATED
	STUDY001-SUBJ0085	2	VOMITING	Severe	RELATED
		5	INSOMNIA	Severe	NOT RELATED
		4	FEVER	Moderate	NOT RELATED
		1	COUGH	Moderate	NOT RELATED
	STUDY001-SUBJ0088	1	HEADACHE	Mild	RELATED

Treatment Arm	Unique Subject ID	AE Seq	Standardized Term (MedDRA)	Severity	Causality
	STUDY001-SUBJ0090	1	FEVER	Severe	RELATED
	STUDY001-SUBJ0091	3	HEADACHE	Mild	NOT RELATED
		5	INSOMNIA	Mild	RELATED
		2	VOMITING	Mild	RELATED
		1	DIARRHEA	Moderate	NOT RELATED
	STUDY001-SUBJ0094	3	NAUSEA	Severe	RELATED
		1	FATIGUE	Severe	NOT RELATED
	STUDY001-SUBJ0095	1	DIZZINESS	Moderate	RELATED
	STUDY001-SUBJ0100	4	PAIN	Moderate	NOT RELATED
		2	PAIN	Mild	NOT RELATED
		3	FEVER	Mild	RELATED
		1	INSOMNIA	Mild	NOT RELATED
Treatment 2	STUDY001-SUBJ0001	5	HEADACHE	Severe	RELATED
		1	NAUSEA	Severe	NOT RELATED
		3	FEVER	Moderate	RELATED
		6	FATIGUE	Mild	NOT RELATED
		4	VOMITING	Moderate	NOT RELATED
	STUDY001-SUBJ0002	1	PAIN	Mild	RELATED
		2	COUGH	Mild	NOT RELATED
		4	NAUSEA	Mild	NOT RELATED
	STUDY001-SUBJ0003	1	DIZZINESS	Severe	NOT RELATED
		2	HEADACHE	Moderate	NOT RELATED
	STUDY001-SUBJ0005	1	FATIGUE	Moderate	RELATED
		2	INSOMNIA	Mild	NOT RELATED
		4	PAIN	Mild	NOT RELATED
	STUDY001-SUBJ0008	3	FEVER	Mild	NOT RELATED
		1	FEVER	Moderate	NOT RELATED
	STUDY001-SUBJ0011	3	DIZZINESS	Mild	RELATED
		1	DIZZINESS	Moderate	RELATED
	STUDY001-SUBJ0014	3	FEVER	Mild	NOT RELATED
		1	FEVER	Mild	RELATED
	STUDY001-SUBJ0016	1	INSOMNIA	Severe	RELATED
		2	VOMITING	Mild	NOT RELATED
	STUDY001-SUBJ0020	3	DIARRHEA	Moderate	NOT RELATED
		1	FEVER	Mild	RELATED
	STUDY001-SUBJ0021	2	VOMITING	Severe	NOT RELATED
		4	VOMITING	Moderate	NOT RELATED
		1	VOMITING	Mild	NOT RELATED
	STUDY001-SUBJ0022	4	DIARRHEA	Severe	RELATED
		3	COUGH	Severe	RELATED
		1	COUGH	Moderate	RELATED
		5	INSOMNIA	Severe	NOT RELATED
	STUDY001-SUBJ0023	2	VOMITING	Moderate	NOT RELATED
		1	DIARRHEA	Severe	RELATED
	STUDY001-SUBJ0027	3	RASH	Moderate	NOT RELATED
		1	NAUSEA	Severe	NOT RELATED
	STUDY001-SUBJ0030	2	DIARRHEA	Mild	NOT RELATED
		5	FEVER	Severe	RELATED
		3	FATIGUE	Severe	NOT RELATED
		6	FEVER	Moderate	NOT RELATED
		1	COUGH	Severe	RELATED
	STUDY001-SUBJ0037	4	DIARRHEA	Mild	NOT RELATED
		2	FATIGUE	Severe	NOT RELATED
		1	COUGH	Severe	RELATED
	STUDY001-SUBJ0039	2	HEADACHE	Severe	RELATED
		1	COUGH	Severe	RELATED
	STUDY001-SUBJ0040	5	DIZZINESS	Mild	RELATED
		4	DIARRHEA	Severe	RELATED
		3	DIZZINESS	Moderate	RELATED
		1	DIARRHEA	Moderate	RELATED
	STUDY001-SUBJ0041	2	DIARRHEA	Mild	NOT RELATED
		3	RASH	Mild	NOT RELATED
		4	DIARRHEA	Severe	RELATED
		6	PAIN	Severe	NOT RELATED
		1	RASH	Mild	RELATED
	STUDY001-SUBJ0042	2	DIARRHEA	Moderate	NOT RELATED
		1	FEVER	Mild	RELATED
	STUDY001-SUBJ0043	2	HEADACHE	Moderate	NOT RELATED
		1	HEADACHE	Mild	RELATED
	STUDY001-SUBJ0048	1	FATIGUE	Mild	RELATED
	STUDY001-SUBJ0050	3	HEADACHE	Mild	RELATED
		1	DIZZINESS	Moderate	RELATED
	STUDY001-SUBJ0052	3	HEADACHE	Mild	RELATED
		4	INSOMNIA	Severe	RELATED
		2	RASH	Moderate	RELATED
		1	INSOMNIA	Mild	NOT RELATED
	STUDY001-SUBJ0053	2	INSOMNIA	Mild	NOT RELATED
		4	NAUSEA	Severe	RELATED
		1	VOMITING	Severe	NOT RELATED
	STUDY001-SUBJ0054	2	INSOMNIA	Mild	NOT RELATED
		5	RASH	Severe	NOT RELATED
		4	VOMITING	Mild	RELATED
		1	DIZZINESS	Moderate	NOT RELATED
	STUDY001-SUBJ0057	5	HEADACHE	Moderate	RELATED
		6	RASH	Severe	RELATED
		2	VOMITING	Severe	NOT RELATED

Treatment Arm	Unique Subject ID	AE Seq	Standardized Term (MedDRA)	Severity	Causality
		4	PAIN	Moderate	RELATED
		1	HEADACHE	Moderate	RELATED
	STUDY001-SUBJ0058	1	COUGH	Moderate	RELATED
	STUDY001-SUBJ0060	3	COUGH	Moderate	NOT RELATED
		1	DIARRHEA	Moderate	RELATED
	STUDY001-SUBJ0061	6	RASH	Moderate	RELATED
		4	COUGH	Mild	NOT RELATED
		3	PAIN	Mild	RELATED
		5	FATIGUE	Moderate	RELATED
		1	FATIGUE	Moderate	NOT RELATED
	STUDY001-SUBJ0062	3	COUGH	Mild	RELATED
		5	FEVER	Moderate	RELATED
		2	DIARRHEA	Mild	RELATED
		1	HEADACHE	Severe	RELATED
	STUDY001-SUBJ0066	5	PAIN	Severe	NOT RELATED
		2	HEADACHE	Moderate	NOT RELATED
		4	FATIGUE	Moderate	RELATED
		1	DIARRHEA	Severe	NOT RELATED
	STUDY001-SUBJ0068	6	VOMITING	Moderate	RELATED
		2	DIARRHEA	Mild	RELATED
		4	FEVER	Severe	RELATED
		3	VOMITING	Mild	NOT RELATED
		1	INSOMNIA	Moderate	RELATED
	STUDY001-SUBJ0071	1	NAUSEA	Mild	RELATED
	STUDY001-SUBJ0072	2	DIARRHEA	Moderate	NOT RELATED
		3	VOMITING	Severe	RELATED
		5	HEADACHE	Severe	NOT RELATED
		1	NAUSEA	Mild	RELATED
	STUDY001-SUBJ0073	1	FATIGUE	Mild	NOT RELATED
	STUDY001-SUBJ0074	3	PAIN	Severe	NOT RELATED
		1	DIARRHEA	Severe	RELATED
	STUDY001-SUBJ0076	2	NAUSEA	Moderate	RELATED
		5	HEADACHE	Severe	RELATED
		3	FATIGUE	Moderate	RELATED
		1	PAIN	Mild	NOT RELATED
	STUDY001-SUBJ0082	2	FEVER	Mild	NOT RELATED
		4	DIZZINESS	Moderate	NOT RELATED
		1	FEVER	Mild	RELATED
	STUDY001-SUBJ0084	2	HEADACHE	Severe	NOT RELATED
		3	FATIGUE	Severe	NOT RELATED
		1	HEADACHE	Mild	NOT RELATED
	STUDY001-SUBJ0086	1	FEVER	Moderate	NOT RELATED
	STUDY001-SUBJ0087	3	DIZZINESS	Mild	NOT RELATED
		4	HEADACHE	Severe	NOT RELATED
		1	FATIGUE	Mild	NOT RELATED
	STUDY001-SUBJ0089	4	DIARRHEA	Moderate	NOT RELATED
		2	RASH	Moderate	RELATED
		5	HEADACHE	Mild	NOT RELATED
		1	DIZZINESS	Severe	NOT RELATED
	STUDY001-SUBJ0096	2	VOMITING	Moderate	NOT RELATED
		1	DIARRHEA	Severe	NOT RELATED
	STUDY001-SUBJ0098	5	FEVER	Mild	RELATED
		6	NAUSEA	Severe	NOT RELATED
		4	FEVER	Moderate	NOT RELATED
		2	NAUSEA	Severe	NOT RELATED
		1	DIARRHEA	Moderate	RELATED
	STUDY001-SUBJ0099	3	DIARRHEA	Moderate	NOT RELATED
		5	FEVER	Severe	NOT RELATED
		4	PAIN	Mild	RELATED
		1	FEVER	Moderate	NOT RELATED